

Predicting Spotify Track Popularity: Regression Modeling of Audio and Structural Features - A Case Study of Taylor Swift

Kerem Bar

Abstract

Accurate music popularity forecasting minimizes financial risks in digital streaming, yet traditional machine learning models often predict popularity based on audio features in isolation, neglecting conceptual context. To bridge this gap, I develop supervised regression architectures to evaluate the predictive capacity of combined acoustic-conceptual matrices for track popularity (RQ1) and their industry generalizability (RQ2). Utilizing Taylor Swift’s discography ($N = 579$) sourced from the Spotify API, I model track popularity ($Y \in [0, 100]$) against twelve features using an Ordinary Least Squares (OLS) Linear Regression baseline benchmarked against optimized Random Forest and Gradient Boosting tree ensembles. To address severe multicollinearity ($r = 0.80$) between **energy** and **loudness**, **energy** was omitted exclusively from the OLS baseline. Additionally, to eliminate chronological data leakage from multi-version catalog re-recordings, I implement an album-family group out-of-sample partition strategy.

All modeled architectures hit a rigid predictive ceiling on unseen eras (Test RMSE ≈ 10.7 , Test $R^2 < 0$), signaling systemic high-bias underfitting. Diagnostic evaluation reveals that extreme distributional asymmetries and sparse tails within the acoustic feature subset exerted high statistical leverage, forcing the OLS model to construct flawed, rigid global rules. Concurrently, the frameworks heavily over-indexed on macro chronology; **release_year** achieved a +12.20 linear coefficient and captured a $\sim 70\%$ importance weight in tree ensembles. This chronological fixation artificially penalized older versions of sonically identical catalog re-recordings, while in the tree ensembles true standard deviations flatten into static, era-specific predictions. Ultimately,

both research questions yielded a negative conclusion; for future work, I propose stabilizing skewed acoustic signals via distribution-based log transformations alongside replacing the static **release_year** with dynamic external exposure proxies.

1 Introduction

In the contemporary digital streaming marketplace, track popularity directly dictates music industry revenue streams, rendering commercial music production a high financial risk. Despite the commercial necessity for accurate forecasting, traditional models frequently fall into an "Acoustic Isolation Trap," predicting popularity solely from isolated audio features without chronological or structural context. Standard predictive frameworks often treat individual tracks as independent and identically distributed (i.i.d.) observations. This oversimplification fails to account for the broader contextual envelope, the chronological era and structural album environment surrounding each track, thereby limiting the utility of acoustic-only models in real-world industry applications. Consequently, two acoustically homogenous tracks can yield vastly divergent streaming volumes based entirely on their chronological release timing or sequential positioning within an album layout.

This project addresses these methodological and structural gaps by formulating track popularity forecasting as a supervised machine learning regression task. I model the continuous target variable ($Y \in [0, 100]$) against an independent feature matrix (X) that integrates nine raw acoustic variables ("acoustic features") with three structural parameters: **release_year**, **track_number**, and **duration_ms** ("conceptual features").

My research evaluates two central questions:

can supervised machine learning architectures accurately predict individual song popularity based on a combined acoustic-conceptual matrix (RQ1), and to what extent can an artist-specific framework successfully generalize across the broader music industry (RQ2)? To capture extreme stylistic variance across a two-decade span, Taylor Swift’s extensive discography is utilized as the primary analytical case study.

To eliminate severe chronological data leakage from multi-version catalog re-recordings, the pipeline clusters variations into unified `album_families` where each cluster represents a distinct chronological era and executes an out-of-sample group split to withhold entire musical eras from training. To bypass the high-dimensional noise and curse of dimensionality inherent in categorical dummy variables under this split, explicit album identities are proactively replaced with the continuous conceptual features to natively capture the chronological and layout essence of each era. This structural substitution ensures architectural stability while enabling the models to evaluate broader temporal trends to test industry generalizability.

To fulfill these analytical objectives, this study establishes a comparative framework across three supervised regression architectures. An Ordinary Least Squares (OLS) Linear Regression model serves as the interpretable baseline framework to evaluate directional relationships through feature coefficients. This baseline is systematically benchmarked against two advanced, non-parametric tree ensemble models, Random Forest and Gradient Boosting Regressors, to analyze tree-based feature importance utilities and evaluate the trade-offs between linear constraints and non-linear expressive power across an identical validation split. Ultimately, the primary contributions of this work are fourfold: (1) the design of an era-isolated validation pipeline via an out-of-sample album-family grouping strategy to eliminate chronological representation leakage; (2) the abstraction and substitution of categorical album identities with continuous conceptual features to natively capture the chronological and layout essence of each era while bypassing zero-variance training constraints;

(3) the establishment of a rigorous comparative benchmark evaluating parametric linear boundaries against non-parametric tree splitting algorithms across the identical out-of-sample validation partitions; and (4) the construction of a statistical diagnostic framework to isolate and analyze the relative predictive weight of macro-chronological features alongside intrinsic acoustic features.

2 Data & Preprocessing

2.1 Dataset Overview and Variable Specification

This project utilizes Taylor Swift’s discography sourced from the Spotify API via Kaggle (<https://www.kaggle.com/datasets/jarredpriester/taylor-swift-spotify-dataset>). The predictive framework models a continuous target variable, popularity ($Y \in [0, 100]$), against an independent matrix (X) of 12 numeric predictors partitioned into two functional categories: nine intrinsic acoustic features (`acousticness`, `danceability`, `energy`, `instrumentalness`, `liveness`, `loudness`, `speechiness`, `tempo`, and `valence`) and three structural conceptual features, which include `track_number` (album sequential position), `duration_ms` (track length in milliseconds), and `release_year` (calendar year of release).

2.2 Data Preprocessing and Engineering Steps

To prepare the raw dataset for stable predictive modeling, the following sequential engineering steps were performed:

Exploratory Outlier Detection and Invalid Data Isolation: During the exploratory analysis phase, an Interquartile Range (IQR) calculation was executed on the popularity metric within the raw dataset (582 observations) to systematically detect statistical anomalies. The mathematical lower fence was established at 7.5. Isolating the data points falling below this threshold revealed exactly three extreme outliers, each carrying a popularity score of 0 (Figure 1). A targeted inspection confirmed these records were non-musical voice memos.

These entries represent invalid data for a predictive framework designed strictly to model commercial musical tracks. To stabilize modeling variance and eliminate structural noise, a deterministic filter (`popularity > 0`) was implemented to remove these 3 records, establishing a finalized clean modeling space of 579 samples.

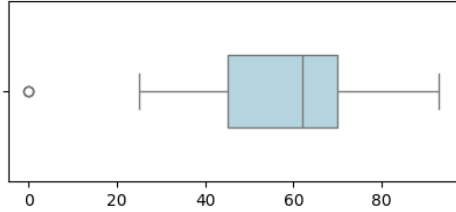


Figure 1: Boxplot Outlier Isolation for Target Variable Popularity

Temporal Feature Extraction: The raw chronological metadata column, `release_date`, is formatted as a text string. Regression estimators and tree splitters cannot compute coefficients or gradients on text objects. Therefore, a slicing operation was executed to extract the four-digit calendar year, creating a new continuous numeric integer feature column designated as `release_year`.

Redundant Resolution (Album Clustering): The raw dataset contains several different versions of the same album under the column `album`, causing identical musical compositions to appear multiple times across the dataset. To resolve this redundancy, a new categorical feature named `album_families` was engineered. This feature explicitly replaced the individual album names by mapping and clustering all corresponding variations into unified, high-level album groups.

Data Partitioning via Group Splitting: To prevent data leakage, data splitting was executed after feature engineering using a group split strategy based on the `album_families` feature with an 80/20 train-test allocation. This group-based partitioning ensures that individual songs, which share identical acoustic footprints, belonging to different versions of the same album, do not simultaneously appear across both the training and testing partitions. This split completely isolated an independent test holdout set spanning three specific album families, which are permanently defined

as the designated target test partition (X_{test}): *1989* (comprising the 2014 original and 2023 re-recorded version), *folklore* (2020), and *evermore* (comprising the 2020 original and 2021 deluxe version).

Feature Standardization: Following data partitioning, feature standardization was applied to all independent variables within the feature matrix. Linear regression models are highly sensitive to the scale of input features. Unscaled predictors with wide numerical boundaries (such as `duration_ms` spanning hundreds of thousands of units) can artificially distort model coefficients over bounded variables operating on smaller scales. To resolve this scale imbalance, Z-score standardization was fitted strictly on the independent training matrix (X_{train}) and subsequently applied to transform the designated test matrix (X_{test}), guaranteeing zero data leakage across all modeled attributes.

2.3 Descriptive Statistics and Feature Distributions

Following data cleaning and feature engineering, baseline summary statistics and mathematical topologies were computed across the unscaled modeling space ($N = 579$) (Figure 2).

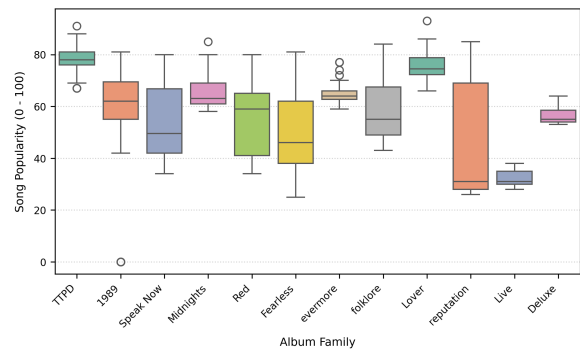


Figure 2: Song Popularity Distribution by Album Families

Table 1 outlines the mathematical boundaries, central tendencies, and empirical distribution shapes verified via skewness testing for all operational variables.

2.4 Challenges and Data Limitations

The finalized feature space presents distinct structural constraints and distributional asym-

Table 1: Summary Statistics of the Finalized Dataset ($N = 579$)

Variable Name	Min - Max	Mean $\pm$ SD	Median (50%)	Distribution / Shape
Popularity (Y)	25.00 - 93.00	58.16 $\pm$ 15.65	62.00	Normal
Release Year	2006.00 - 2024.00	2017.70 $\pm$ 5.43	2020.00	Left Skewed
Track Number	1.00 - 46.00	11.39 $\pm$ 8.05	10.00	Right Skewed
Duration (ms)	83,253 - 613,026	240,469.53 $\pm$ 45,477.47	235,766.00	Right Skewed
Acousticness	0.00 - 0.97	0.33 $\pm$ 0.33	0.18	Right Skewed
Danceability	0.18 - 0.90	0.58 $\pm$ 0.11	0.59	Normal
Energy	0.12 - 0.95	0.57 $\pm$ 0.19	0.57	Normal
Instrumentalness	0.00 - 0.33	0.00 $\pm$ 0.03	0.00	Right Skewed
Liveness	0.03 - 0.93	0.16 $\pm$ 0.13	0.11	Right Skewed
Loudness	-15.91 - -1.93	-7.62 $\pm$ 2.84	-7.34	Normal
Speechiness	0.02 - 0.52	0.05 $\pm$ 0.05	0.04	Right Skewed
Tempo	68.10 - 208.92	122.47 $\pm$ 30.43	119.06	Normal
Valence	0.04 - 0.94	0.39 $\pm$ 0.20	0.37	Normal

metries that introduce specific challenges for predictive modeling frameworks:

The Categorical Dummy Partitioning Constraint: Incorporating individual album identities via binary dummy variables is structurally incompatible with a group-based data splitting strategy. The corresponding dummy indicator column of the target test partition contains only static zero values throughout the entire training dataset (X_{train}). Linear estimators cannot mathematically calculate coefficients for features with zero training variance, and tree-based algorithms cannot isolate valid splits on empty arrays. To ensure cross-partition architectural stability, the framework bypasses individual categorical dummies and instead relies on continuous numeric conceptual features, to capture temporal and layout configurations natively.

High Feature Multicollinearity: Empirical evaluation via the training correlation matrix reveals a severe linear dependency between core acoustic predictors, notably a high positive correlation between **energy** and **loudness** ($r = 0.80$). This structural redundancy presents a mathematical limitation for standard ordinary least squares (OLS) linear estimators, as highly collinear features destabilize coefficient weights. To mitigate this limitation, the acoustic features are treated adaptively across different frameworks: **energy** is excluded from the final baseline linear model to restore structural stability but is safely retained within the tree-based architectures which naturally tolerate collinear inputs.

Distributional Asymmetry and Non-Normality: As outlined in the summary statistics (Table 1), several predictors deviate significantly from symmetric normal distributions, creating modeling constraints for linear estimators. Features capturing specialized audio traits (**speechiness**, **instrumentalness**)

and layout dimensions (**track_number**, **duration_ms**) exhibit heavy right-skewness, where the bulk of the data clusters at lower bounds and is pulled by isolated extended values. Conversely, **release_year** displays a severe left-skewed trajectory, heavily weighted toward recent highly prolific production eras. These unscaled asymmetries can act as statistical leverage that distorts linear regression weights.

3 Methodology and Predictive Models

3.1 Baseline Linear Regression Architecture

To establish an interpretable predictive baseline, this study utilizes a standard linear regression model, which serves as the definitive parametric benchmark for continuous estimation tasks. This architecture modeling framework maps the target variable (y_i) as a parametric linear combination of the input predictors. Formally, the architecture estimates the continuous popularity score using the following formulation:

$$y_i = \beta_0 + \sum_{j=1}^p \beta_j x_{ij} + \epsilon_i \quad (1)$$

Where β_0 represents the intercept, β_j denotes the static directional coefficients calculated via ordinary least squares minimizing residual sum of squares, and ϵ_i represents the stochastic error term. This parametric structure allows for direct empirical validation of how individual attributes dictate popularity.

To resolve the severe multicollinearity between **energy** and **loudness** ($r = 0.80$) documented in Section 2.4, the linear framework evaluates two distinct configurations. The first is the Linear Regression (Full) configuration, which retains the complete 12-predictor feature matrix to diagnose collinearity damage. The second is the Linear Regression (Stabilized) configuration, which intentionally excludes the **energy** feature to restore coefficient stability. This stabilized framework serves as the definitive baseline model benchmarked against the subsequent tree architectures.

3.2 Improved Non-Parametric Tree Ensembles

To transcend linear constraints and map non-monotonic distribution profiles, the baseline is benchmarked against Random Forest and Gradient Boosting Regressors. This tree-based ensemble approach relies on hierarchical branch splitting to partition the continuous feature space into homogenous regions rather than fitting a global linear slope. Formally, both ensemble estimators construct an additive predictive function composed of M individual base decision trees:

$$y_i = \sum_{m=1}^M f_m(x_i) \quad (2)$$

Where each $f_m(x_i)$ represents an independent regression tree mapping local regional splits. In Random Forest, the M learners are trained completely in parallel via bootstrap aggregation to minimize variance, whereas Gradient Boosting optimization fits trees sequentially to minimize squared loss gradients computed from prior steps. This branch splitting logic naturally accommodates collinear features, allowing the safe reintegration of **energy** alongside **loudness** without risking statistical destabilization.

For both frameworks, the total number of estimators was held constant ($M = 100$). Hyperparameter tuning targeted tree depth regularization, executing an experimental grid search sweeping the maximum tree depth boundary from 1 to 10 on the training matrix to identify the optimal complexity limit that balances training performance with test partition generalization.

4 Evaluation and Results

4.1 Evaluation Setup and Performance Metrics

To evaluate architectural generalization under realistic baseline constraints, the framework utilizes the out-of-sample train/test group partition strategy detailed in Section 2.2, bypassing standard cross-validation to prevent catalog representation leakage. Model performance is quantified using two continuous regression pa-

rameters: Root Mean Squared Error (RMSE) and the Coefficient of Determination (R^2).

The continuous optimization parameters are formulated and evaluated based on the following mathematical frameworks:

Root Mean Squared Error (RMSE): Calculates the square root of the average squared differences between predictions and ground-truth values. This metric provides a pragmatic diagnostic tool because it expresses the prediction error directly in the original units of the target variable (Spotify popularity points), allowing for intuitive evaluation of forecasting variance.

Coefficient of Determination (R^2): Measures the proportion of variance in the dependent target variable that is predictable from the independent feature matrix, formulated as:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3)$$

Where y_i represents the actual popularity score for track i , $\hat{y}_i$ denotes the predicted popularity score calculated by the model, and $\bar{y}$ represents the mean popularity score of the sample.

Theoretically, R^2 is bounded from above at 1.0 for perfect predictions, but it can yield negative baseline outputs. In this study, R^2 is leveraged as a critical architectural diagnostic framework, where a negative value mathematically indicates that the residual sum of squares (the numerator) exceeds the total sum of squares around the sample mean (the denominator). Such an output demonstrates that the trained models' structural features perform worse than a naive horizontal benchmark that simply applies the mean popularity score ($\bar{y}$) to every unseen track, signaling severe architectural underfitting.

4.2 Quantitative Empirical Results

The Linear Regression (Full) configuration evaluated all 12 acoustic and conceptual variables to establish an initial predictive baseline without any feature exclusion. This full architecture achieved a Train RMSE of 11.4240 and a Train R^2 of 0.5595. When evaluated on the unseen test partition (X_{test}), this setup yielded a Test RMSE of 10.8603 and a Test R^2 of -0.1536.

In comparison, the Linear Regression (Stabilized) configuration excluded the **energy** feature to resolve the multicollinearity issues doc-

umented in Section 2.4. This configuration achieved a Train RMSE of 11.4302 and a Train R^2 of 0.5590, while yielding an optimized Test RMSE of 10.7815 and a Test R^2 of -0.1369 on the unseen test partition. Comparing these two setups demonstrates that omitting **energy** mitigated out-of-sample prediction errors, making the stabilized framework the preferred baseline model despite both configurations yielding negative Test R^2 scores.

To optimize both tree-based models, a hyperparameter grid search swept the maximum tree depth parameter from 1 to 10. Evaluating this sweep isolated a uniform maximum depth of 4 as the global empirical threshold where out-of-sample performance peaked for both architectures. To justify this constraint, increasing the depth to 10 caused the Gradient Boosting model to heavily over-memorize the training data, reaching a Train RMSE of 1.9103 and a Train R^2 of 0.9877, while its corresponding Test R^2 collapsed to -0.2648. Therefore, this maximum depth boundary was strictly enforced across both tree frameworks to safeguard against this overfitting trajectory.

The Random Forest ensemble model achieved a Train RMSE of 7.6028 and a Train R^2 of 0.8049. On the unseen test partition, this model yielded a Test RMSE of 10.7317 and a Test R^2 of -0.1264.

The Gradient Boosting ensemble model achieved a Train RMSE of 3.2835 and a Train R^2 of 0.9636, representing the most successful training fit across all primary experiments. On the unseen test partition, this model achieved a Test RMSE of 10.7190 and a Test R^2 of -0.1238.

In summary, although the tree-based models achieved significantly stronger training performance than the linear baselines, with Gradient Boosting demonstrating the highest overall training fit, all primary models ultimately collapsed during testing. Regardless of architectural complexity, the out-of-sample metrics converged to a Test RMSE ≈ 10.7 and a negative R^2 , indicating a shared structural limitation in the engineered feature space (Table 2).

Table 2: Comprehensive Empirical Summary of Model Performance

Model Architecture	Feature Configuration	Train RMSE	Train R^2	Test RMSE	Test R^2
Linear Regression (Full)	Full 12 Features	11.4240	0.5595	10.8603	-0.1536
Linear Regression (Stabilized)	11 Features (Excludes energy)	11.4302	0.5590	10.7815	-0.1369
Random Forest Ensemble	Full 12 Features	7.6028	0.8049	10.7317	-0.1264
Gradient Boosting Ensemble	Full 12 Features	3.2835	0.9636	10.7190	-0.1238

4.3 Linear Parameter Coefficient Analysis and Diagnosis

To diagnose the out-of-sample breakdown of the baseline linear model, its learned structural coefficients must be evaluated against the empirical distributions of the dataset ($N = 579$) as detailed in Table 1 (Summary Statistics of the Finalized Dataset). While popularity follows an approximately normal curve (Mean=58.16, SD=15.65), **acousticness** (Mean=0.33, SD=0.33) and **liveness** (Mean=0.16, SD=0.13) exhibit severe right-skewed asymmetry. During training, these sparse right-hand tails exert high statistical leverage, forcing the model to construct rigid acoustic rules based on global parameters as visualized in Figure 3: a fixed positive coefficient for acoustic density (+1.2395) and a fixed negative coefficient for live performance signatures (-0.7238).

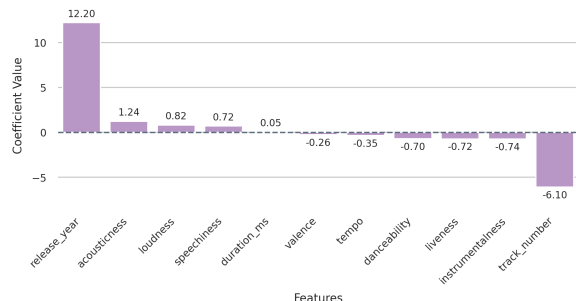


Figure 3: Standardized Coefficient Weights

To test these rules objectively, a data-driven One Standard Deviation (1σ) Rule (Mean + SD) was implemented to isolate the extreme tails of the test partition (X_{test}). As shown in Table 1, this mathematically defines an empirical **acousticness** tail threshold at > 0.70 and a **liveness** tail threshold at > 0.30 . On the normally distributed target variable, high popularity is defined above 73.81, and low popularity sits below 42.51. Stricter 2σ boundaries were rejected as they exceed physical data limits, yielding empty sub-partitions.

When evaluated out-of-sample, these rigid

parameters collapse because a uniform acoustic formula for popularity does not exist across a multi-genre catalog. Under the positive Acousticness Rule, the model falsely equates high acoustic density with guaranteed success; however, the test data reveals internal polarization where frontline acoustic hits achieve peak traction (**popularity**=77–78 at **acousticness**=0.77–0.92) while deeper acoustic cuts carrying near-absolute signatures (**acousticness**=0.964) collapse to low commercial performance (**popularity**=43).

Concurrently, the negative Liveness Rule misinterprets maximalist studio pop productions; due to layered arena-style production and stadium echoes, these studio tracks register elevated liveness values (**liveness**=0.32–0.38) yet command peak commercial success (**popularity**=71–80). Conversely, actual live sessions yield extreme liveness (**liveness**=0.791) alongside low platform traction (**popularity**=54). Failing to differentiate between high-traction studio engineering and lower-performing deep cuts or raw live variants, the global linear framework accumulates severe, compounding directional errors.

This blind application of false acoustic rules is further compounded by the model’s over-indexing on macro chronology, driven by a massive positive coefficient of +12.2020 assigned to **release_year**. This multiplier creates severe structural anomalies inside the test partition (X_{test}) by artificially penalizing the 2014 tracks of *1989* against its 2023 iteration and similarly splitting *evermore*’s 2020 tracks from its 2021 release.

Ultimately, because the baseline model applies rigid assumptions to skewed training tails and blindly tracks calendar dates, its out-of-sample prediction errors are massive. The Residual Sum of Squares (SS_{res}) strictly exceeds the Total Sum of Squares (SS_{tot}), causing the Test R^2 to collapse below zero ($R^2 < 0$), confirming that a global linear framework performs worse than a naive baseline tracking the sample mean.

4.4 Tree Ensemble Diagnostic and Variance Collapse Analysis

Both tree ensemble architectures fall into an extreme chronological trap that severely lim-

its their predictive performance. Rather than mapping localized acoustic structures, the tree-splitting hierarchies allow the **release_year** feature to completely dominate architectural utility, capturing 70.4% importance in the Random Forest model and 69.6% in the Gradient Boosting machine (Figure 4). All remaining features are heavily marginalized, with individual attributes failing to clear an 8% utility threshold (Figure 4).

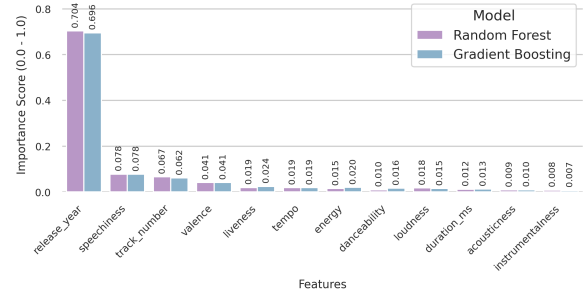


Figure 4: Feature Importance of Both Tree Ensemble Models

Both algorithms rely so heavily on this single temporal feature to execute their top-level branches, that they effectively operate as macro historical binning systems rather than generalized musical estimators, mirroring the linear baseline’s penalty on older, sonically identical tracks. This structural blindness triggers a total Predictive Variance Collapse, flattening true empirical variations into static, era-specific predictions (Figure 5). The genuine popularity standard deviations of *folklore* (2020) ($\sigma = 11.23$), *evermore* (2020) ($\sigma = 5.44$), *evermore* (2021) ($\sigma = 3.13$), and *1989* (2023) ($\sigma = 5.96$) are compressed to near-static predicted spreads of 0.53, 0.46, 0.45, and 1.24 in Random Forest, and 2.71, 2.60, 2.46, and 3.29 in Gradient Boosting, respectively (Figure 5).

Ultimately, expanding algorithmic complexity to hierarchical tree ensembles fails to resolve this predictive ceiling. Mirroring the baseline linear framework, out-of-sample performance collapses below zero ($R^2 < 0$) across both models. This convergence demonstrates empirically that despite their non-linear expressive power, the tree architectures cannot compensate for a structurally flawed feature matrix, still performing worse than a naive baseline tracking the sample mean.

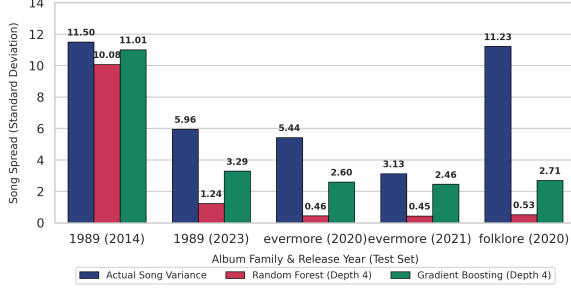


Figure 5: Empirical Evidence of Predictive Variance Collapse

5 Conclusions and Future Work

5.1 Conclusions

This study evaluated two central research questions regarding the capacity of supervised machine learning regression architectures to forecast music track popularity (RQ1) and their structural generalizability across the broader music industry (RQ2). Based on the empirical results and diagnostic evaluations, both research questions yield a definitive negative conclusion.

Utilizing Taylor Swift’s multi-genre discography operated as a structural double-edged sword. First, her extreme stylistic shifts generated heavy distributional asymmetries and long tails in core acoustic metrics (Table 1), which forced the baseline linear model to construct flawed, rigid global rules over these statistical levers. Second, the training catalog presented conflicting internal album family popularity profiles that defied uniform acoustic patterns (Figure 2), highlighted by the tightly compressed, high-popularity tier of *TTPD* album_family (median ≈ 80), contrasted against the massive vertical dispersion and lower median of the *reputation* album_family (median ≈ 30).

Faced with these conflicting acoustic profiles during training, both parametric and non-parametric algorithms largely discounted individual musical attributes. Instead, they over-indexed on `release_year` as a macro chronological shortcut, evaluating tracks based on their historical release windows rather than their individual sonic characteristics (Figure 3 and Figure 4). When evaluated out-of-sample,

this shared chronological fixation led to a universal algorithmic collapse, where all modeled architectures uniformly converged to negative R^2 scores ($R^2 < 0$). Specifically for the tree ensemble architectures, this temporal shortcut triggered the Predictive Variance Collapse (Figure 5), where the estimators flattened true empirical standard deviations into static, era-specific predictions. This behavior definitively indicates that in addressing RQ1, individual song popularity cannot be predicted using acoustic-conceptual feature matrices.

In addressing RQ2, these predictive frameworks cannot successfully generalize to other artists across the broader music industry. The learned logic is strictly dependent on the unique longevity, atypical catalog replication milestones, and specific historical streaming timeline of Taylor Swift’s career. The models function as a specialized career tracker rather than a generalized musical estimator. Ultimately, this project demonstrates that neither baseline linear models nor expanded algorithmic complexity can rescue a framework when the engineered features evaluate complex artistic progression through a rigid chronological lens.

5.2 Future Work

To overcome the predictive and structural boundaries identified in this study while preserving Taylor Swift’s core musical blueprint, future research should implement two practical methodological adjustments:

Distribution-Based Preprocessing: Retaining acoustic variables that exhibit symmetric, normal distributions in their original form, while applying a standard $\log(x+1)$ transformation exclusively to highly skewed, non-normal metrics to properly accommodate their zero-value bounds.

Chronological Abstraction via External Proxies: Replacing the static, highly artist-specific `release_year` feature with dynamic, external exposure metrics (e.g., media appearance frequencies, social media volume, and touring schedules).